

BHAVESH GURNANI

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EDUCATION

- **Indian Institute of Technology Delhi (IIT Delhi)** *Expected May 2026*
 - B.Tech and M.Tech in Computer Science & Engineering (Dual Degree) | CGPA: 8.50/10.0
- **Lord Buddha Public School, Kota** *Apr 2021 (CBSE XII), Apr 2019 (CBSE X)*
 - Class XII (PCM): 84% | Class X: 94%

SCHOLASTIC ACHIEVEMENTS

- **JEE Advanced 2021:** All India Rank **116** (out of approx. 200,000 candidates)
- **JEE Mains 2021:** All India Rank **493** (out of >1,000,000 candidates); **100% Percentile** in Mathematics
- **KVPY Fellowship 2020 (IISc):** All India Rank **305** (SA Stream)
- **NTSE Scholar 2019 (NCERT):** Ranked among Top **2000** students nationwide
- **INMO Qualifier:** Ranked among Top **300** students nationwide

EXTRACURRICULAR ACTIVITIES

- Achieved **Top 50** selection **twice** in the **Tower Research Limestone Data Challenge** for offline sessions
- Led **Inter IIT Tech Meet** for Adobe Research problem; designed solution for artifacts in **AI-generated images**
- Took **ML in Astronomy Workshop**; built **LSTM-Attention models** on light curves for object classification[Code]

EXPERIENCE

- **Teaching Assistant** | *Indian Institute of Technology Delhi / New Delhi* *Jul 2025 – Present*
 - *COL333 (Artificial Intelligence)* under Prof. Mausam; created quizzes/assignments, and evaluated scripts.
- **Quantitative Researcher Intern** | *Ebullient Securities / Gurugram* *May 2025 – July 2025*
 - Built and backtested alpha signals for short-term equity directionality from equities and futures.
 - Designed and live-tested a custom trading strategy in C++ using top alphas (40% annualised, Calmar 12).
 - Automated alpha-generation pipeline using LLMs with custom evaluation loops for iterative refinement.
- **Software/Research Engineer Intern** | *CompilerAI Labs Private Limited / New Delhi* *May 2024 – July 2024*
 - Project under the guidance of Prof. Sorav Bansal
 - Implemented MISRA C static analysis rules within Clang compiler across Preprocessor, AST, and LLVM stages.
 - Enhanced Equivalence Checker tool for formal verification of compiled executable code against C source code.

PUBLICATIONS

- **ProbMedTOD: A Probabilistic Task-Oriented Dialogue System for Patient History Taking (In Review)**
Authors: Vishal Vivek Saley, **Bhavesh Gurnani**, Dinesh Raghu, Mausam
 - Introduced **ProbMedTOD**, integrating LLMs with Bayesian Networks for patient history-taking.
 - Designed patient simulator and dialogue-level metric; achieved 21-pt MRR gain over baselines.

SELECTED PROJECTS

- **Reasoning using Diffusion Language Models** | *MTP under Prof. Parag Singla* *June 2025 – Present*
 - Implemented Path Planning with Remasking strategies for LLaDa on datasets like sudoku and ARC-AGI.
 - Improved decoding strategy using custom Remasking Strategy boosting 9×9 Sudoku accuracy from 75.7% to 96.1%.
- **Improving Dialogue Systems for Medical Diagnosis** | *BTP under Prof. Mausam* *July 2024 – May 2025*
 - Modeled large-scale medical text data as Bayesian Networks to capture causal disease-symptom relationships.
 - Designed novel inference techniques for Bayesian Networks, improving performance over standard RAG systems.
- **Object-Centric Learning for Scene Decomposition** | *DL Course Final Project* *Jan 2024 – Apr 2024*
 - Implemented Slot Attention on CLEVRText and designed slot-conditioned diffusion model with custom U-Net.
 - Benchmarked performance using ARI metric and achieved interpretable object-centric scene representations.
- **Intelligent Bot for Alternating Markov Games** | *AI Course Final Project* *Aug 2023 – Dec 2023*
 - Developed AI agents for strategic games using depth-limited Minimax with domain heuristics optimized via RL.
 - Implemented DQN enhanced with MCTS inspired by Expert Iteration, achieving superior gameplay performance.
- **Lexer, Parser, Interpreter for Language with Rationals** | *Prof. S. Arun Kumar* *Jan 2023 – Apr 2023*
 - Built complete compiler with custom BigInt/Rational modules, lexer using ML-Lex, and parser with ML-Yacc.
 - Implemented interpreter with static scoping supporting variables, conditionals, loops, and recursion.
- **Custom MultiThreading Library** | *Course Project under Prof. Abhilash Jindal* *Feb 2023 – Mar 2023*
 - Designed a user-level multithreading library in C with round-robin scheduling and synchronization primitives.
 - Benchmarked against POSIX pthread, demonstrating efficient scheduling with reduced overhead.